Amir Mirzai Golpayegani

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Summary

M.Sc. Computer Science graduate in Calgary with experience building Python/C++ systems for geospatial data infrastructure, machine learning, and large-scale Earth observation workflows. Skilled in PyTorch, spatial data pipelines, model evaluation, DGGS-based storage, and backend APIs, with applied research experience through a BigGeo/Mitacs collaboration.

Education

M.Sc. in Computer Science GPA: **3.94**/4.0 Sep 2022 – Feb 2026
University of Calgary *Calgary, Canada*

B.Sc. in Computer Engineering GPA: **3.69**/4.0 Sep 2016 – Feb 2021
Sharif University of Technology *Tehran, Iran*

Technical Skills

Programming: Python, C++, JavaScript, SQL
ML / AI: PyTorch, Scikit-learn, TensorFlow, CNNs, model evaluation, LLM workflows, AI agents
Geospatial / Data: NumPy, Pandas, SciPy, Google Earth Engine API, Discrete Global Grid System (DGGS), GDAL, Rasterio, SNAP
Backend / Engineering: Django, Node.js, REST APIs, PostgreSQL, Docker, Git, Linux, AWS, GitHub Actions
Visualization / Automation: Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

Work Experience

Research Software Developer Sep 2022 – Feb 2026
BigGeo / Mitacs, University of Calgary *Calgary, Canada*

- Built C++/Python software for scalable processing, storage, and retrieval of Earth observation data.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGS-based encoding pipelines to convert satellite imagery into structured tensor files.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Reduced storage by 38% for scaled Canada-wide data with 1,112 GeoTIFF tiles and 16K+ DGGS tensors.

Back-End Developer Jun 2020 – Jul 2022
Asr Gooyesh Pardaz *Tehran, Iran*

- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

Deep Learning Teaching Assistant Jul 2026
Neuromatch Academy *Remote, United States*

- Mentored an international pod of ∼15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

Teaching Assistant Dec 2022 – Apr 2025
Department of Computer Science, University of Calgary *Calgary, Canada*

- **Developing Big Data Applications**: Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale**: Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems**: Assisted +75 students with SQL, database design, course projects, grading, and exam support.

Research and Technical Projects

Spatiotemporal Earth Observation Data System Using DGGS Sep 2022 – Feb 2026
First-author publication | M.Sc. thesis | C++, Python, DGGS, Wavelets, B-splines, Time-Series

- Designed a DGGS-based framework for real-time multiresolution management of satellite imagery.
- Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.
- Implemented a triangular wavelet scheme to support multiresolution retrieval without image-pyramid redundancy.

- Built B-spline temporal approximation to compress time-series data and retrieve missing timestamps locally.
- Retrieved 105,489 DGGS cells per time-series query in 13.9s, compared with about 220s using global retrieval.

DEM Super-Resolution Framework May 2024 – Feb 2026
Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization

- Trained deep ResNet models for 4× DEM super-resolution, reconstructing 30m elevation data from 120m inputs.
- Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and systematically collect and preprocess mountainous terrain data worldwide.
- Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors.
- Developed a Polyscope-based 3D visualization tool to render DEM tiles from multiple angles and compare reconstruction quality in 3D.

Agentic Job Search Automation Platform May 2026 – Present
OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API

- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.
- Designed LLM-based workflows for resume/job matching, skill-gap analysis, and application decision support.
- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.
- Prototyped the initial workflow in n8n before migrating toward a more flexible OpenClaw-based agent system.

Publication

Amir Mirzai Golpayegani, Mahmudul Hasan, Faramarz F. Samavati. *Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGS*. Remote Sensing, 2025. doi:10.3390/rs17040570

Certificates

Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford University, Coursera Oct 2024
Applied Machine Learning in Python, University of Michigan, Coursera Jul 2020

Languages

English: Full Professional Proficiency (TOEFL **115**/120)
Persian: Native Proficiency

Achievement

Teaching Assistant Excellence Laureate 2024/2025
Department of Computer Science, University of Calgary — awarded for outstanding teaching support (*link*)

Ranked 17th Nationwide in Iran University Entrance Exam (Konkour) May 2016
Top 0.001% among all participants